

FedMuon: Federated Learning with Bias-Corrected LMO-Based Optimization

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Abstract

This report presents a reproducibility study of FedMuon, a federated learning algorithm that adapts the Muon optimizer; a linear minimization oracle (LMO)-based method to the federated setting. Vanilla Muon fails in federated learning because the LMO operator is biased: the average of individually orthogonalized client updates does not equal the orthogonalization of the averaged update. FedMuon corrects this bias by introducing control variates that align local momentum estimates with the global gradient direction before applying the LMO. We reproduced the core experimental results from Takezawa et al. [1] on FashionMNIST using a LeNet-5 model across 16 clients under both homogeneous (Dirichlet $\beta = 10$) and heterogeneous ($\beta = 0.1$) data distributions. Our experiments confirm that FedMuon consistently outperforms FedAvg with SGD, FedAvg with Adam, SCAFFOLD, and Local Muon, particularly under high data heterogeneity. We further investigate the effect of Newton–Schulz iteration count on convergence quality.

1. Introduction and Background

Federated learning (FL) is a distributed machine learning paradigm in which a shared global model is trained across a large number of clients, each holding private local data that never leaves the device. In each communication round, selected clients perform local gradient-based updates, and a central server aggregates the resulting weight changes into a new global model. This architecture is appealing for privacy sensitive applications, but introduces several optimization challenges, chief among them being *data heterogeneity*: in real deployments, client datasets often follow very different distributions, causing local update directions to diverge; a phenomenon known as *client drift*.

Muon [4] is a recently proposed optimizer that has garnered attention for its strong empirical performance in centralized training settings. The core idea of Muon is to apply Nesterov momentum and then orthogonalize the resulting momentum matrix via the Newton–Schulz algorithm before taking a gradient step. This operation is a form of linear minimization oracle (LMO) over the spectral norm ball, and it effectively normalizes gradient directions across layers, leading to faster convergence than Adam [5] on standard benchmarks. However, adapting Muon to the federated setting is non-trivial because the LMO is a highly nonlinear operator, which creates a fundamental incompatibility with FedAvg-style aggregation.

The paper *FedMuon: Federated Learning with Bias-corrected LMO-based Optimization* [1] identifies this incompatibility and proposes a principled fix. This report is a reproducibility study of that work.

2. Motivation

Three interconnected observations motivate the development of FedMuon.

Muon is faster than Adam in centralized training. Muon consistently reaches the same validation accuracy

as Adam with fewer gradient steps and fewer epochs, suggesting that LMO-based updates have favorable geometry for neural network optimization.

Communication efficiency is critical in federated learning. Every communication round in FL requires transmitting all model parameters between clients and the server. Reducing the number of rounds needed to reach a target accuracy directly translates to lower bandwidth consumption, reduced energy usage on mobile devices, and cheaper deployments. If Muon’s convergence advantage carries over to FL, it could meaningfully reduce communication cost.

Existing federated optimizers exhibit accuracy ceilings under strong data heterogeneity. FedAvg with SGD or Adam frequently saturates at sub-optimal accuracy when client data distributions are highly non-IID (e.g., Dirichlet $\beta = 0.1$). A qualitatively different optimizer family, based on spectral geometry rather than per-parameter adaptivity, may escape these plateaus.

3. Problem Formulation

3.1 Federated Optimization Setting

We consider the standard federated optimization problem:

$$\min_{\mathbf{x}} F(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n F_i(\mathbf{x}), \quad (1)$$

where F_i is the local loss on client i and $\mathbf{x}$ are the shared model parameters. There are $n = 16$ clients in total, of which $S = 8$ are randomly selected each round. Each selected client performs $K = 5$ local gradient steps before sending updates to the server.

Data heterogeneity is controlled via a Dirichlet distribution with concentration parameter β . A small β (e.g., 0.1) yields highly heterogeneous client datasets where each client holds data mostly from one or two classes; a large β (e.g., 10.0) approximates the IID setting.

3.2 Why Vanilla Muon Fails in Federated Learning

In centralized training, Muon maintains a momentum matrix $\mathbf{M}$ and updates it each step as:

$$\mathbf{M} \leftarrow (1 - \alpha)\mathbf{M} + \alpha \nabla F(\mathbf{x}). \quad (2)$$

The parameter update is then $\mathbf{x} \leftarrow \mathbf{x} + \eta \cdot \text{lmo}(\mathbf{M})$, where $\text{lmo}(\mathbf{M})$ denotes the orthogonalized momentum matrix (the polar factor of $\mathbf{M}$), computed via the Newton–Schulz iteration.

When naively applied in federated learning, each client i maintains its own local momentum $\mathbf{M}_i$ and independently applies the LMO. After K local steps, the server averages the resulting updates. The core problem is that the LMO is a *nonlinear* operator. It discards all singular values and retains only the singular vectors. As a consequence, the average of the LMO outputs does not equal the LMO of the average:

$$\frac{1}{n} \sum_{i=1}^n \text{lmo}(\mathbf{M}_i^{(r+1)}) \neq \text{lmo}\left(\frac{1}{n} \sum_{i=1}^n \mathbf{M}_i^{(r+1)}\right). \quad (3)$$

This inequality means that averaging locally orthogonalized updates introduces a systematic bias: each client’s LMO is well-calibrated to its own local gradient distribution, but the resulting average does not correctly represent the global gradient direction. In practice, this causes training to diverge or accuracy to collapse, particularly under data heterogeneity.

4. Method: FedMuon Algorithm

4.1 Bias Correction via Control Variates

FedMuon addresses the bias problem in Eq. (3) by applying the LMO not to the raw local momentum $\mathbf{M}_i$ but to a bias-corrected version of it. The correction uses two control variates:

- $\mathbf{C}_i$: a *local* control variate stored on each client, initialized to zero and updated each round. It is a snapshot of client i ’s momentum from the previous communication round.
- $\mathbf{C}$: a *global* control variate maintained at the server, computed as the average of all clients’ $\mathbf{C}_i$ values.

The key idea is that $\mathbf{C}_i$ captures client i ’s local momentum bias from the previous round, while $\mathbf{C}$ represents the global average of that bias. By computing $\mathbf{M}_i - \mathbf{C}_i + \mathbf{C}$ before applying the LMO, each client’s local update is shifted toward the global gradient direction. This de-biasing ensures that the average of $\text{lmo}(\mathbf{M}_i - \mathbf{C}_i + \mathbf{C})$ approximates lmo applied to the globally averaged momentum.

4.2 Algorithm Description

The FedMuon local update rule for each client i at communication round r , local step k , is given in Algorithm 1, alongside Local Muon for comparison.

Algorithm 1 FedMuon — Local Update on Client i , Round r

Require: Global model $\mathbf{X}_i^{(r,0)}$, global control variate $\mathbf{C}^{(r)}$, local control variate $\mathbf{C}_i^{(r)}$, learning rate η , momentum α , local steps K

- 1: **for** $k = 0, 1, \dots, K - 1$ **do**
- 2: $\mathbf{M}_i^{(r,k+1)} \leftarrow (1 - \alpha)\mathbf{M}_i^{(r,k)} + \alpha \nabla F_i(\mathbf{X}_i^{(r,k)}; \xi_i^{(r,k)})$
- 3: $\mathbf{X}_i^{(r,k+1)} \leftarrow \mathbf{X}_i^{(r,k)} + \eta \cdot \text{lmo}(\mathbf{M}_i^{(r,k+1)} - \mathbf{C}_i^{(r)} + \mathbf{C}^{(r)})$
- 4: **end for**
- 5: Send $\Delta \mathbf{X}_i = \mathbf{X}_i^{(r,K)} - \mathbf{X}_i^{(r,0)}$ to server
- 6: Update local control variate: $\mathbf{C}_i^{(r+1)} \leftarrow \mathbf{M}_i^{(r,K)}$

After K local steps, each client sends its model update to the server. The server aggregates updates via FedAvg averaging, then computes the updated global control variate $\mathbf{C}^{(r+1)}$ as the average of all clients’ updated local control variates. This is then broadcast back to all participating clients for the next round.

Contrasting with Local Muon (the naive baseline), which applies $\text{lmo}(\mathbf{M}_i^{(r,k+1)})$ directly without any correction, FedMuon’s correction term ensures that the LMO receives an input that accounts for the discrepancy between local and global gradient directions.

4.3 Newton–Schulz LMO Approximation

Computing the exact polar factor of a matrix (i.e., the exact LMO) requires a full singular value decomposition, which is expensive. The Newton–Schulz iteration provides an efficient polynomial approximation that converges quadratically. With T iterations of Newton–Schulz, denoted lmo_T , the approximation quality improves as T increases. FedMuon is theoretically shown to converge for any $T \geq 0$, and to converge faster as T grows, making it practical to tune T for the desired accuracy efficiency tradeoff.

5. Experimental Setup

We reproduced experiments on FashionMNIST, a 10-class image classification dataset consisting of 60,000 training and 10,000 test grayscale images of clothing items. The model is LeNet-5, a standard convolutional network with two convolutional layers and three fully connected layers, well-suited for this benchmark.

The federated setup uses $n = 16$ clients with $S = 8$ selected per round, $K = 5$ local gradient steps per round, a batch size of 32, and training for 1,000 communication rounds (2,000 rounds for the Newton–Schulz ablation). Data partitioning follows a Dirichlet distribution with $\beta \in \{0.1, 10.0\}$.

We compare FedMuon against the following baselines: FedAvg [2] with SGD, FedAvg with Adam, SCAFFOLD [3] with SGD, SCAFFOLD with Adam, and Local Muon (vanilla FedAvg with Muon as the local optimizer, without bias correction). SCAFFOLD is a strong variance-reduction baseline designed specifically to address client drift.

For the ablation on LMO approximation quality, we vary the Newton–Schulz iteration count $T \in \{0, 1, 2, 4\}$ and train for 2,000 rounds under both $\beta = 0.1$ and $\beta = 10.0$.

Table 1: Summary of experimental hyperparameters.

Hyperparameter	Value
Dataset	FashionMNIST (60k train / 10k test)
Model	LeNet-5
Total clients n	16
Selected per round S	8
Local steps K	5
Batch size	32
Communication rounds	1,000 (2,000 for ablation)
Dirichlet β	0.1 (heterogeneous), 10.0 (homogeneous)
Newton–Schulz T	$\{0, 1, 2, 4\}$ (ablation)

6. Results

6.1 Heterogeneous Data ($\beta = 0.1$)

Under strong data heterogeneity, FedMuon achieves the highest final test accuracy, reaching approximately 89–90% on FashionMNIST after 1,000 communication rounds. FedAvg with SGD and SCAFFOLD with Adam both plateau near 85–87%. SCAFFOLD with SGD achieves peaks lower than the FedAvg-Adam

variants and is seen to have higher variance. Critically, Local Muon (vanilla Muon without bias correction) stalls near 65–68%, confirming the theoretical prediction that the LMO bias causes divergent client updates and ultimately an accuracy collapse under high heterogeneity.

These results validate the core claim of the paper: the bias correction in FedMuon is essential, not incidental. Without it, applying Muon in federated learning is actively harmful relative to simpler baselines.

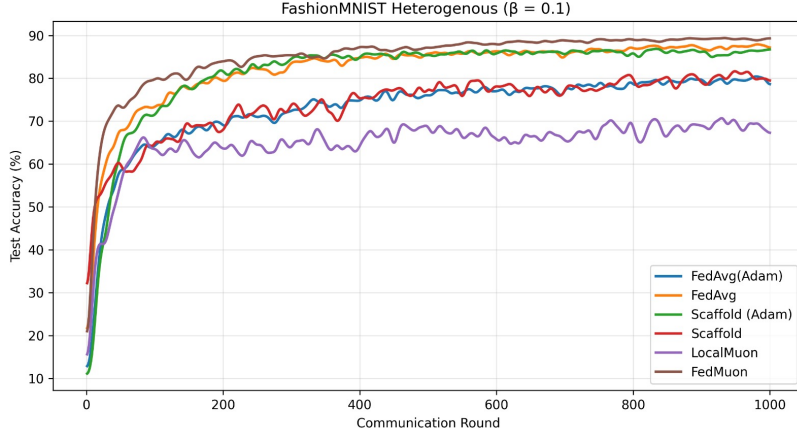


Figure 1: Test accuracy on FashionMNIST over 1,000 communication rounds under heterogeneous data ($\beta = 0.1$). FedMuon (brown) converges fastest and achieves the highest accuracy. Local Muon (purple) stalls near 65–68% due to LMO bias.

6.2 Homogeneous Data ($\beta = 10.0$)

When data distributions are approximately IID across clients, the bias correction matters less. In this setting, FedMuon and Local Muon converge to comparable final accuracies, both reaching approximately 90%. Most methods achieve more than 80% accuracy within 100 rounds under homogeneous conditions. This result is expected: with homogeneous data, local gradient directions are already aligned, so the correction term $\mathbf{C}_i - \mathbf{C}$ is small and the two variants behave similarly.

FedMuon still attains the highest peak accuracy, indicating that even under IID conditions, the bias correction does not hurt performance and may provide marginal gains in early convergence.

In addition, we extended our testing to models with different parameters $\beta = 0.1$ (0.5 and 5) to test performance under different data heterogeneity settings and observed similar trends.

6.3 Effect of Newton–Schulz Iterations (T)

The ablation on T reveals a consistent monotone improvement with increasing iteration count. Under heterogeneous conditions ($\beta = 0.1$), FedMuon with $T = 0$ (no LMO approximation, equivalent to using raw momentum without orthogonalization) still converges but to a substantially lower final accuracy of around 80–82%. $T = 1$ improves to roughly 86%, $T = 2$ reaches approximately 88–90%, and $T = 4$ achieves the highest accuracy near 90%.

Importantly, the performance gap between $T = 2$ and $T = 4$ is small, while the gap between $T = 0$ and $T = 1$ is large. This suggests diminishing returns from additional Newton–Schulz iterations beyond $T = 2$,

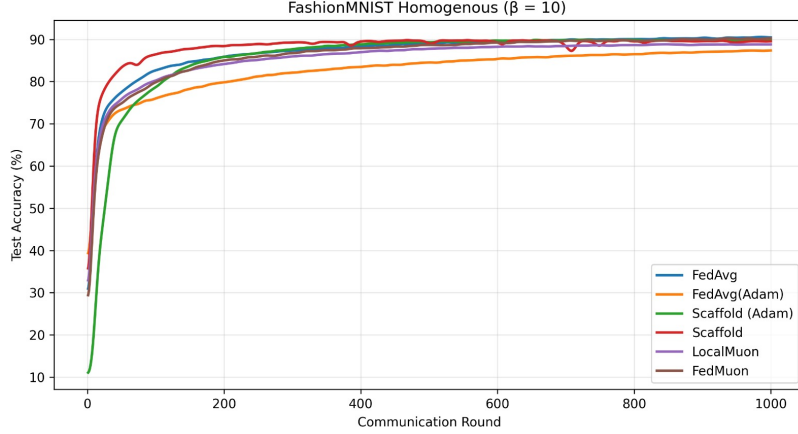


Figure 2: Test accuracy on FashionMNIST over 1,000 communication rounds under homogeneous data ($\beta = 10.0$). FedMuon and Local Muon converge comparably; most methods exceed 80% within 100 rounds.

making $T = 2$ a practical sweet spot that balances computational cost and convergence quality.

Under homogeneous conditions ($\beta = 10$), the performance differences across T values are compressed, with $T = 4$ again performing best. Since the performance gap is relatively small, increasing T does not incur a significant trade-off, making additional iterations a low-cost way to achieve marginal gains.

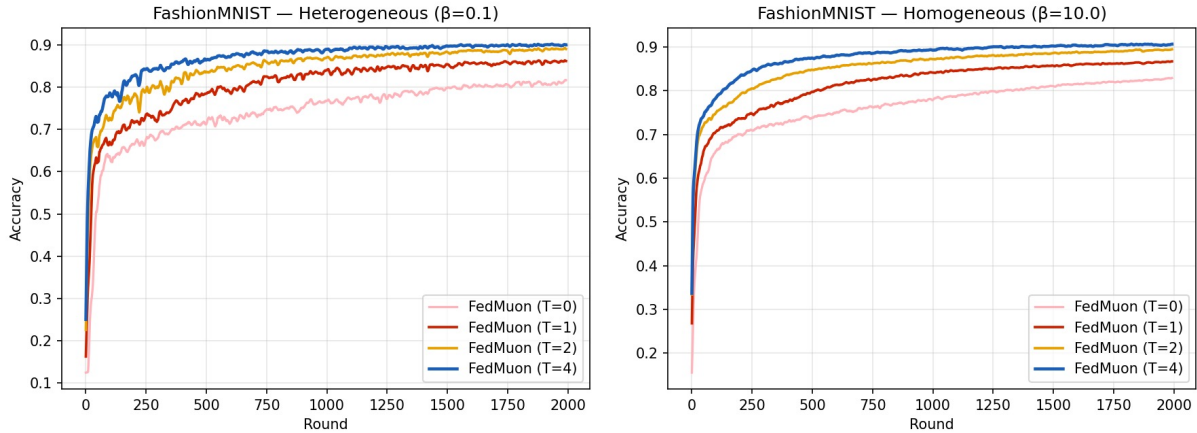


Figure 3: Impact of Newton-Schulz iterations T on FedMuon performance over 2,000 rounds. Left: heterogeneous ($\beta = 0.1$). Right: homogeneous ($\beta = 10.0$). Gains diminish beyond $T = 2$.

7. Take-Home Messages

This reproducibility study supports three key takeaways from the original FedMuon paper.

The LMO bias is a real and significant obstacle. Naively extending Muon to federated learning without bias correction yields performance far below Adam, the current standard. The control variate mechanism in FedMuon is a targeted and effective fix.

Data heterogeneity amplifies the importance of bias correction. The performance gap between FedMuon and Local Muon grows dramatically as β decreases. Under IID conditions, both converge similarly; under non-IID conditions, only FedMuon maintains competitive accuracy.

Newton–Schulz iteration count is a practical tuning knob. Setting $T = 2$ provides a strong tradeoff between computation and accuracy. $T = 0$ still converges (a non-trivial result, since LMO quality is theoretically necessary for the analysis), while $T = 4$ provides marginal gains over $T = 2$.

The broader implication is that LMO-based optimization, which has shown promise in centralized training, can be extended to federated learning with a relatively lightweight modification. FedMuon opens a direction for future work on spectral-norm-ball constrained federated optimizers, potentially enabling faster convergence for privacy-preserving distributed training at scale.

8. Team Member Contributions

This project was carried out collaboratively by three team members. The primary contributions of each member are outlined below.

Samprithi Suresh Premalatha: Led the implementation and evaluation of the FedMuon and Local Muon algorithms. Additionally, contributed to the development and analysis of Newton–Schulz (NS) iterations, as well as to the overall project design and integration.

Krishna Chemburkar: Led the implementation and evaluation of the SCAFFOLD and SCAFFOLD-Adam algorithms. Also contributed to the development and analysis of Newton–Schulz (NS) iterations, and to the overall project design and integration.

Hira Nayak: Led the implementation and evaluation of the FedAvg and FedAvg-Adam algorithms. Also contributed to the development and analysis of Newton–Schulz (NS) iterations, and to the overall project design and integration.

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